APO-NAPRO-Na
APO-NAPRO-Na DS

Naproxen Sodium Tablets USP 275mg and 550mg
Therapeutic Classification:
Analgesic, anti-inflammatory agent

Actions and Clinical Pharmacology: Naproxen sodium has demonstrated analgesic, anti-inflammatory and antipyretic properties in human clinical studies and in classical animal test systems. It exhibits an anti-inflammatory effect even in adrenalectomized animals and therefore its action is not mediated through the pituitary-adrenal axis. It is not a corticosteroid. It inhibits prostaglandin synthetase, as do certain other nonsteroidal analgesic/anti-inflammatory agents. As with other agents, however, the exact mechanism of its anti-inflammatory and analgesic actions is not known. Blood loss and gastroscopy studies with normal volunteers showed that daily administration of 1100 mg of naproxen sodium caused significantly less gastric bleeding and erosion than 3250mg of ASA. At the recommended dosage, the analgesic effect of naproxen sodium was shown to be comparable to that observed using 650mg of ASA. The analgesic effect is obtained within one hour and can last at least 7 hours.

Pharmacokinetics: Naproxen sodium is freely soluble in water and is completely absorbed from the gastrointestinal tract. Plasma levels are obtained in patients within 20 minutes and peak levels in 1 hour. It is extensively bound to plasma protein and has a plasma half-life of approximately 13 hours. The preferred route of excretion is via the urine with only 1% of the dose excreted in the feces. A bioavailability study was performed using normal human volunteers. The rate and extent of absorption of naproxen after a single oral 550mg dose of Anaprox 275mg and Apo-Napro-Na 275mg tablets was measured and compared. The results are summarized as follows:

Parameter	Anaprox 275mg	Apo-Napro-Na 275mg	% Diff.
AUC 0-32 (mcg.hrs/mL)	857.3	834.9	-2.6
Cmax (mcg/mL)	83.0	81.2	-2.2
Tmax (hrs)	1.3	0.9	-32.9
$t^{1}/_{2}$ (hrs)	14.0	14.2	+1.4

Indications and clinical use: APO-NAPRO-Na or APO-NAPRO-Na DS (naproxen sodium) is indicated for the relief of mild to moderately severe pain accompanied by inflammation in conditions such as musculo-skeletal trauma and post-dental extraction. It is also indicated for the relief of pain associated with post-partum cramping and dysmenorrhea.

Contraindications: Naproxen sodium is contraindicated in patients with peptic ulcer or active inflammatory diseases of the gastrointestinal system. Known or suspected hypersensitivity to the drug. Naproxen sodium should not be used in patients in whom acute asthmatic attacks, urticaria, rhinitis or other allergic manifestations are precipitated by ASA or other nonsteroidal anti-inflammatory agents. Fatal anaphylactoid reactions have occurred in such individuals.

Warnings: Peptic ulceration, perforation and gastrointestinal bleeding, sometimes severe and occasionally fatal have been reported during therapy with nonsteroidal anti-inflammatory drugs (NSAIDs) including naproxen sodium. APO-NAPRO-Na and APO-NAPRO-Na DS (naproxen sodium) should be given under close medical supervision to patients prone to gastrointestinal tract irritation particularly those with a history of peptic ulcer, diverticulosis or other inflammatory disease of the gastrointestinal tract. In these cases the physician must weigh the benefits of treatment against the possible hazards. Patients taking any NSAID including this drug should be instructed to contact a physician immediately if they experience symptoms or signs suggestive of peptic ulceration or gastrointestinal bleeding. These reactions can occur without warning symptoms or signs and at any time during the treatment. Elderly, frail and debilitated patients appear to be at higher risk from a variety of adverse reactions from nonsteroidal anti-inflammatory drugs (NSAIDs). For such patients, consideration should be given to a starting dose lower than usual, with individual adjustment when necessary and under close supervision. See 'PRECAUTIONS' for further advice.

Use in Pregnant and Lactating Women: The safety of this drug in pregnancy and lactation has not been established and its use during these events is therefore not recommended. Reproduction studies have been performed in rats, rabbits and mice. In rats, pregnancy was prolonged when naproxen sodium was given before the onset of labour; when it was given after the delivery process had begun, labour was protracted. Similar results have been found with other nonsteroidal anti-inflammatory agents and the evidence suggests that this may be due to decreased uterine contractility resulting from the inhibition of prostaglandin synthesis. Moreover, because of the known effect of drugs of this class on the human fetal cardiovascular system (closure of ductus arteriosus), use during late pregnancy should be avoided. The naproxen anion readily crosses the placental barrier. It has been found in the milk of lactating women at a concentration approximately 1% of that found in the plasma.

WARNINGS
RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID
Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.
Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Precautions: APO-NAPRO-Na or APO-NAPRO-Na DS (naproxen sodium) should not be used concomitantly with the related drug naproxen since they both circulate in plasma as the naproxen anion.
Use in the Elderly Patient: One study indicates that after the administration of naproxen, although total plasma concentration of naproxen is unchanged, the unbound plasma fraction of naproxen is increased in the elderly. The implication of this finding for naproxen sodium dosing is unknown, but caution is advised when high doses are required. As with other drugs used in the elderly, it is prudent to use the lowest effective dose.
Gastrointestinal System: If peptic ulceration is suspected or confirmed, or if gastrointestinal bleeding or perforation occurs APO-NAPRO-Na or APO-NAPRO-Na DS should be discontinued, an appropriate treatment instituted and the patient closely monitored. There is no definitive evidence that the concomitant administration of histamine H2- receptor antagonists and/or antacids will either prevent the occurrence of gastrointestinal side effects or allow continuation of naproxen sodium therapy when and if these adverse reactions appear. APO-NAPRO-Na or APO-NAPRO-Na DS should be given under close supervision to patients prone to gastrointestinal tract irritation, in patients with a history of peptic ulcer or in patients with diverticulosis. Gastrointestinal bleeding, sometimes severe and occasionally fatal and peptic ulcer have occurred in patients receiving naproxen.
Renal Function: As with other nonsteroidal anti-inflammatory drugs, long term administration of naproxen sodium to animals has resulted in renal papillary necrosis and other abnormal renal pathology. In humans, there have been reports of acute interstitial nephritis with hematuria, proteinuria, and occasionally nephrotic syndrome. A second form of renal toxicity has been seen in patients with prerenal conditions leading to the reduction in renal blood flow or blood volume, where the renal prostaglandins have a supportive role in the maintenance of renal perfusion. In these patients, administration of a nonsteroidal anti-inflammatory drug may cause a dose-dependent reduction in prostaglandin formation and may precipitate overt renal decompensation. Patients at greatest risk of this reaction are those with impaired renal function, extracellular volume depletion, sodium restrictions, heart failure, liver dysfunction, those taking diuretics and the elderly. Discontinuation of nonsteroidal anti-inflammatory therapy is typically followed by recovery to the pretreatment state. Naproxen sodium and its metabolites are eliminated primarily by the kidneys, therefore, the drug should be used with great caution in patients with significantly impaired renal function. In these cases lower doses of naproxen sodium should be anticipated and patients carefully monitored. During long term therapy, kidney function should be monitored periodically. In clinical trials a few patients developed mild elevations in BUN. The significance of this is unknown. Naproxen sodium should not be used chronically in patients having baseline creatinine less than 20mL/minute.
Hepatic Function: As with other nonsteroidal anti-inflammatory drugs, borderline elevations of one or more liver tests may occur in up to 15% of patients. These abnormalities may progress, may remain essentially unchanged or may be transient with continued therapy. A patient with symptoms and/or signs suggesting liver dysfunction, or in whom an abnormal liver test has occurred, should be evaluated for evidence of the development of more severe hepatic reaction while on therapy with this drug. Severe hepatic reactions including jaundice and cases of fatal hepatitis have been reported with this drug as with other nonsteroidal anti-inflammatory drugs. Although such reactions are rare, if abnormal liver tests persist or worsen, if clinical signs and symptoms consistent with liver disease develop, of if systemic manifestations occur (e.g. eosinophilia, rash, etc), this drug should be discontinued. During long-term therapy, liver function tests should be monitored periodically. If this drug is to be used in the presence of impaired liver function, it must be done under strict observation. Chronic alcoholic liver disease and probably also other forms of cirrhosis reduce the total plasma concentration of naproxen, but the plasma concentration of unbound naproxen is increased. The implication of this finding for naproxen sodium dosing is unknown, but caution is advised when high doses are required. It is prudent to use the lowest effective dose.
Fluid and Electrolyte Balance: Fluid retention and edema have been observed in patients treated with naproxen sodium. Therefore, as with many other nonsteroidal anti-inflammatory drugs, the possibility of precipitating congestive heart failure in elderly patients or those with compromised cardiac used with caution in patients with heart failure, hypertension or other conditions predisposing to fluid retention. Serum electrolytes should be monitored periodically during long term therapy, especially in those patients at risk. Each APO-NAPRO-Na tablet contains approximately 25mg of sodium and each APO-NAPRO-Na DS tablet contains approximately 50mg of sodium. This should be considered in patients whose overall intake of sodium must be markedly restricted. Although sodium retention has not been reported in metabolic studies, the drug should be used with caution in patients with fluid retention, hypertension or heart failure.
Hematology: Drugs inhibiting prostaglandin biosynthesis do interfere with platelet function to some degree; therefore, patients who may be adversely affected by such an action should be carefully observed when naproxen sodium is administered. Blood dyscrasias associated with the use of nonsteroidal anti-inflammatory drugs are rare, but could be associated with severe consequences. Patients with initial hemoglobin values of 10 grams or less who are to receive long term therapy should

have hemoglobin values determined frequently.
Infection: The anti-inflammatory, antipyretic and analgesic effects of naproxen sodium may mask the usual signs of infection and the physician should be alert for development of infection in patients receiving APO-NAPRO-Na or APO-NAPRO-Na DS.
Ophthalmology: Because of adverse eye findings in animal studies with drugs of this class, it is recommended that ophthalmic studies be carried out within a reasonable period of time after starting therapy and at periodic intervals thereafter if the drug is to be used for an extended period of time.
Central Nervous System: Caution should be exercised by patients whose activities require alertness if they experience drowsiness, dizziness, vertigo or depression during therapy with the drug.
Hypersensitivity Reactions: Anaphylactoid reactions to naproxen or naproxen sodium, whether of the true allergic type or the pharmacologic idiosyncratic (e.g. aspirin syndrome) type, usually but not always occur in patients with a known history of such reactions. Therefore, carefully questioning of patients for such things as asthma, nasal polyps, urticaria, and hypotension associated with nonsteroidal anti-inflammatory drugs before starting therapy is important. In addition, if such symptoms occur during therapy, treatment should be discontinued.
Cardiovascular Function: It is possible that patients with questionable or compromised cardiac function may be at a greater risk when taking naproxen sodium.
Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose and duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.
There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSIAD use.
Hypertension: NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired antihypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.
Heart Failure: Fluid retention and oedema have been observed in some patients taking NSAIDs, there caution is advised in patients with fluid retention or heart failure.
Gastrointestinal Events: All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patient about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.
Severe Skin Reactions: NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.
Use in Children: APO-NAPRO-Na and APO-NAPRO-Na DS are not recommended in children under 16 years of age because the safety and dose schedule have not been established in this age group.
Laboratory Tests: APO-NAPRO-Na and APO-NAPRO-Na DS decrease platelet aggregation and prolong bleeding time. This effect should be kept in mind when bleeding times are determined. The administration of naproxen sodium may result in increased urinary values of 17 ketogenic steroids because of an interaction between the drug and/or its metabolites with m-dinitrobenzene used in this assay. Although 17 hydroxycorticosteroid measurements (Porter-Silber test) do not appear to be artifactually altered, it is suggested that naproxen therapy be temporarily discontinued 48 hours before adrenal function tests are performed. The drug may interfere with some urinary assays of 5 hydroxy indoleacetic acid (5HIAA).

Drug Interactions: The naproxen anion may displace from their binding sites other drugs which are also albumin-bound and may lead to drug interactions. For example, in patients receiving bishydroxycoumarin or warfarin, the addition of naproxen sodium could prolong the prothrombin time. These patients should, therefore, be under careful observation. Similarly, patients receiving naproxen sodium and a hydantoin, sulfonamide or sulfonylurea should be observed for signs of toxicity to these drugs.
The natriuretic effect of furosemide has been reported to be inhibited by some drugs of this class. Inhibition of renal lithium clearance leading to increases in plasma lithium concentrations have also been reported. Naproxen sodium and other nonsteroidal anti-inflammatory drugs can reduce the antihypertensive effect of propranolol and other beta blockers.
The rate of absorption of naproxen sodium is altered by concomitant administration of antacids but is not adversely influenced by the presence of food. Probenecid given concurrently increases naproxen anion plasma levels and extends its plasma half-life significantly.
Caution is advised in the concomitant administration of naproxen sodium and methotrexate since naproxen and other nonsteroidal anti-inflammatory agents have been reported to reduce the tubular secretion of methotrexate in an animal model, thereby possibly enhancing its toxicity.

Adverse Reactions: The most common adverse reactions encountered with nonsteroidal anti-inflammatory drugs are gastrointestinal, of which peptic ulcer, with or without bleeding, is the most severe. Fatalities have occurred on occasion, particularly in the elderly.
Adverse reactions reported in controlled clinical trials are listed below:
1) Denotes incidence of reported reaction between 3% and 9%.
2) Denotes incidence of reported reactions between 1% and 3%. Reactions occurring in less than 1% of the patients during controlled clinical trials and through voluntary reports since marketing are unmarked.

Gastrointestinal: Heartburn (1), constipation (1), abdominal pain (1), nausea (1), diarrhea (2), dyspepsia (2), stomatitis (2), diverticulitis (2), gastrointestinal bleeding, hematemesis, melena, peptic ulceration with or without bleeding and/or perforation, vomiting, ulcerative stomatitis. Central Nervous System: Headache (1), dizziness (1), drowsiness (1), lightheadedness (2), vertigo (2), depression (2) and fatigue (2). Occasionally patients had to discontinue treatment because of the severity of some of these complaints (headache and dizziness). Other adverse effects were inability to concentrate, malaise, myalgia, insomnia and cognitive dysfunction (i.e decreased attention span, loss of short-term memory, difficulty with calculations). Dermatologic: Pruritus (1), ecchymoses (1), skin eruptions (1), sweating (2), purpura (2), alopecia, urticaria, skin rash, erythema multiforme, Stevens-Johnson syndrome, epidermal necrolysis, photosensitive dermatitis, exfoliative dermatitis and erythema nodosum. Cardiovascular Reactions: Dyspnea (1), peripheral edema (1), palpitations (2), congestive heart failure and vasculitis. Special Senses: Tinnitus (1), hearing disturbances (2), hearing impairment and visual disturbances. Hematologic: Eosinophilia, granulocytopenia, leukopenia, thrombocytopenia, agranulocytosis, aplastic anemia and hemolytic anemia. Renal: Glomerular nephritis, hematuria, interstitial nephritis, nephrotic syndrome, nephropathy and tubular necrosis. Hepatic Changes: Abnormal liver function tests, jaundice, cholestasis and hepatitis. Others: Thirst (2), muscle weakness, anaphylactoid reactions, menstrual disorder, pyrexia (chills and fever), angioneurotic edema, hyperglycemia, hypoglycemia, hematuria, hepatitis and eosinophilic pneumonitis.

Symptoms and Treatment of Overdosage: Significant overdosage may be characterized by drowsiness, heartburn, indigestion, nausea or vomiting. No evidence of toxicity or late sequelae have been reported 5 to 15 months after ingestion for three to seven days of doses up to 3000mg of naproxen. One patient ingested a single dose of 25g naproxen and experienced mild nausea and indigestion. It is not known what dose of the drug would be life threatening. The oral LD50 of the drug is 543mg/kg in rats, 1234mg/kg in mice, 4110 mg/kg in hamsters and greater than 1000 mg/kg in dogs.
Should a patient ingest a large number of naproxen sodium tablets, the stomach may be emptied and usual supportive measures employed. Animal studies suggest that the prompt administration of 5 grams of activated charcoal would tend to reduce markedly the absorption of the drug. In dogs, 0.5g/kg of charcoal was effective in reducing the plasma levels of naproxen when given after the drug. Hemodialysis does not decrease the plasma concentration of naproxen because of the high degree of its protein binding. However, hemodialysis may still be appropriate in the management of renal failure.

Dosage and Administration: After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortness possible duration should be used.
The recommended starting dose of APO-NAPRO-Na (naproxen sodium) for adults is two 275mg tablets followed by one 275mg tablet every 6 to 8 hours, as required. The total daily dose should not exceed 5 tablets (1375mg). Alternatively, one APO-NAPRO-Na DS tablet (550mg naproxen sodium) given twice daily may be used.

Availability of Dosage Forms:
Apo-Naproxen-Na 275mg
Each oval, biconvex, blue, film-coated tablets, engraved “APO-275” on one side and plain on the other, contains 275mg of naproxen sodium. Available in PVC Blisters of 500s.
Apo-Naproxen-Na 550mg
Each oval, biconvex, blue, film-coated tablets, engraved “APO-550” on one side and plain on the other, contains 550mg of naproxen sodium. Available in PVC Blisters of 500s.

Storage Conditions: Store below 25°C. Protect from light. Protect from moisture.

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